

AAYUSH SINGH

ComputerScienceEngineering Student

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Summary

Computer Science Engineering student at VIT-AP University with hands-on experience in Java, Spring Boot, MySQL, and RESTful API development. Built secure backend applications using Spring Security and JWT Authentication. Passionate about backend development, software engineering, and problem solving through Data Structures and Algorithms.

Projects

Personal Blog API

- Developed a RESTful blog platform using Java, Spring Boot, Spring Security, and MySQL.
- Implemented JWT-based authentication and role-based authorization for secure access control.
- Designed and developed 15+ REST API endpoints for users, posts, comments, and authentication.
- Applied layered architecture using Controllers, Services, Repositories, and DTOs.
- Performed API testing using Postman and integrated global exception handling for robust error responses.

Smart Chessboard

- Developing an IoT-based chessboard using ESP32, Hall Effect sensors, and WS2812B LEDs.
- Designing a system capable of tracking all 64 chess squares through sensor-based piece detection.
- Implementing real-time move validation and LED-based highlighting of legal moves.
- Building a scalable architecture for future Bluetooth, Wi-Fi, and mobile application integration.

Skills

- Programming Languages: Java, SQL
- Backend Development: Spring Boot, Spring Security, RESTful APIs, JWT Authentication, Authentication & Authorization
- Database: MySQL, Database Design, SQL Queries
- Tools: Git, GitHub, Maven, Postman, DataGrip
- Core Concepts: Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), Exception Handling, Dependency Injection, REST Architecture
- Development Practices: API Testing, Version Control, Debugging

Education

- VIT-AP University 2024 – 2028
Bachelor of Technology (B.Tech) in Computer Science Engineering
Amaravati, Andhra Pradesh

Achievements

- Solved 100+ Data Structures and Algorithms problems on LeetCode.
- Participated in Smart India Hackathon (SIH) 2025.
- Built backend applications using Spring Boot and MySQL.